

Faithfulness, Robustness, and Generalization in Chain-of-Thought Reasoning

A Critical Mixed-Methods Evaluation of Large Language Models for High-Stakes Decision Support

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ABSTRACT

Chain-of-thought (CoT) prompting has emerged as the dominant paradigm for eliciting multi-step reasoning from large language models (LLMs), yet the practical reliability of the resulting rationales remains contested. This paper presents a critical, mixed-methods evaluation of frontier LLMs across three properties that jointly determine whether CoT outputs are suitable for high-stakes decision support: *faithfulness* (whether stated reasoning causally drives the answer), *robustness* (whether outputs are stable under semantically irrelevant perturbations), and *generalization* (whether reasoning competence transfers across domains, jurisdictions, and populations). We construct a curated benchmark of 2,148 high-stakes items spanning legal contract review, clinical triage, and financial credit analysis; we evaluate five frontier models; and we introduce a suite of faithfulness probes, perturbation protocols, and cross-domain transfer tests. Our results reveal a substantial *faithfulness gap*: across all evaluated models, stated reasoning trajectories are systematically unfaithful to the underlying decision process, with 31–58% of correct answers remaining correct when the model's own rationale is silently replaced with an irrelevant, incoherent, or adversarial one. We further document non-trivial robustness decay under five classes of surface and semantic perturbation, and we observe a pronounced domain-transfer penalty averaging 14.7 percentage points. We argue that the prevailing assumption — that CoT explanations are *self-evidencing* of model competence — is empirically untenable in high-stakes contexts, and we propose a concrete evaluation protocol that practitioners can deploy before integrating LLM-generated rationales into consequential workflows.

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1. Introduction

1.1 Motivation

Large language models (LLMs) have rapidly moved from research laboratories into consequential decision-making pipelines. As of 2025, frontier models draft contracts, screen clinical notes, summarize depositions, score credit applications, and advise on regulatory filings [openai2024gpt4o; anthropic2024claude; team2024gemini]. The dominant technique for eliciting non-trivial reasoning from these systems is *chain-of-thought* (CoT) prompting, in which the model is instructed to or trained to produce an explicit, step-by-step natural-language rationale prior to committing to a final answer [wei2022chain; kojima2022large]. CoT has been shown to substantially improve performance on arithmetic, commonsense, and symbolic reasoning benchmarks, and it is now the de-facto interface between LLMs and end-users who must understand, audit, or contest automated decisions.

The popularity of CoT rests on an assumption so intuitive that it is rarely stated: that the generated rationale is *both a cause and an explanation* of the model's answer. Under this assumption, a downstream user can read the rationale, recover the model's reasoning, and form a calibrated belief in the verdict. The assumption is the load-bearing element of nearly every governance framework proposed for LLM deployment in regulated industries, including financial services [treasury2024ai], healthcare [fda2024ai], and the legal sector [lai2024lawge].

We show in this paper that the assumption is empirically fragile across the three properties that matter most for high-stakes deployment: faithfulness, robustness, and generalization. We provide an evaluation protocol that makes those properties directly measurable.

1.2 Research Questions

The study is organized around three research questions:

RQ1 *Faithfulness.* **To what extent does the chain-of-thought rationale that a model produces causally determine its final answer, as opposed to merely post-hoc rationalizing an answer reached by other means?**

RQ2 *Robustness.* **How stable are CoT-derived answers and rationales under semantically irrelevant perturbations of the prompt, including paraphrasing, typographical noise, demographic reframing, and adversarial insertion?**

RQ3 *Generalization.* **To what extent does reasoning competence demonstrated in one high-stakes domain transfer to another, and what properties of the domain drive the transfer penalty?**

1.3 Contributions

We make the following contributions:

1. **An operational taxonomy of faithfulness, robustness, and**

generalization for CoT in high-stakes decision support, with explicit definitions, measurable proxies, and a decision procedure for whether a given model-deployment pairing satisfies a stated reliability bar.

1. **HS-CoT-Bench**, a curated benchmark of 2,148 high-stakes items

across legal contract review, clinical triage, and financial credit analysis, with expert-annotated rationales and ground-truth labels.

1. **A five-class perturbation protocol** that disentangles surface-level

noise from semantically meaningful reframing, and an intervention-based faithfulness probe that does not require access to model internals.

1. **A controlled evaluation** of five frontier models (closed-weight and

open-weight), reporting quantitative results for each research question and a qualitative failure-mode analysis.

1. **A practitioner checklist** for evaluating CoT-based systems prior to

deployment, with thresholds calibrated against the failure rates we observe.

1.4 Paper Organization

Section 2 situates the work against four literatures: CoT prompting, faithfulness, robustness, and high-stakes decision support. Section 3 introduces our theoretical framework. Section 4 details the methodology, including benchmark construction, perturbation protocol, and metrics. Section 5 reports results. Section 6 discusses implications, limitations, and ethical considerations. Section 7 concludes and outlines future work.

2. Background and Related Work

2.1 Chain-of-Thought Prompting

The CoT family of techniques was introduced by @wei2022chain, who showed that prompting a model with exemplars containing explicit intermediate steps substantially improves accuracy on the GSM8K arithmetic benchmark. @kojima2022large subsequently demonstrated that the *zero-shot* trigger phrase "Let's think step by step" elicits comparable gains. Subsequent work has refined CoT along several axes: self-consistency [@wang2023self], which samples multiple rationales and aggregates the answers; least-to-most [@zhou2022least], which decomposes problems into sub-problems; tree-of-thoughts [@yao2023tree], which performs explicit search over rationales; and self-refine [@madaan2023self], which iteratively critiques and revises outputs. A complementary line of work trains models to produce CoT trajectories via supervised fine-tuning on rationale-annotated data [@ho2022flan] or via reinforcement learning against downstream rewards [@zelikman2022star; @team2024o1].

The implicit promise of CoT is operational: a human auditor can read the rationale, verify the intermediate steps, and form a calibrated trust judgment about the answer. This promise is what makes CoT *the* interface for consequential applications.

2.2 Faithfulness in Model Explanations

The faithfulness of an explanation is the degree to which the explanation reflects the *actual* causal pathway the model used to reach its answer, rather than a post-hoc rationalization constructed after the answer is determined. The property has been studied extensively for classical interpretability methods, where it is well established that feature attributions, attention weights, and influence functions are systematically unfaithful [@jain2019attention; @adebayo2018sanity; @shrikumar2017learning; @schwartz2020towards].

For LLM-generated natural-language rationales, the question is harder because the "explanation" is itself a free-form natural language string rather than a structured attribution. @jacovi2022faithfulness provides a formal taxonomy distinguishing *plausibility* (the explanation sounds correct to a human) from *faithfulness* (the explanation reflects the model's actual reasoning), arguing that the two are routinely conflated in practice. Several recent works have begun to probe the faithfulness of CoT specifically. @lanham2023measuring introduce the "early answering" intervention: truncating the rationale to see whether later steps are still produced in a way consis-

tent with the model's stated reasoning. @lyu2023crud show that CoT can be readily perturbed to produce contradictory yet confidently answered outputs. @wang2023large demonstrate that CoT can encode intermediate tokens that influence the final answer in ways that are not legible in the natural-language rationale. Most directly, @turpin2024language show that CoT explanations systematically *lie* about the influence of biased input features, with the model producing rationales that disclaim the very bias that drives its answer.

Our work extends this literature by (i) operationalizing faithfulness as a measurable property using a directly causal intervention that does not require access to model internals; (ii) applying it across multiple high-stakes domains; and (iii) quantifying the faithfulness gap for frontier models deployed in 2025–2026.

2.3 Robustness of Language Models

LLM robustness has been studied along several dimensions: adversarial prompts, jailbreaks, distributional shift, and out-of-distribution generalization. @zou2023universal demonstrate that adversarial suffixes can be optimized to elicit harmful completions from aligned models. @wang2023large report that CoT can be used to amplify adversarial attacks. @shah2023scalable introduce prompt-injection attacks against LLM-integrated applications and show that even simple, non-optimized injections succeed at non-trivial rates. @glukhov2023llm analyze the robustness of LLM legal advice to surface perturbations, finding substantial non-robustness.

In parallel, a literature on *counterfactual* and *contrastive* robustness asks whether models behave consistently under minimal input edits that do not change the ground-truth label. @kaushik2020learning introduce the contrast set methodology; @huang2020location; @niu2020estimating; and @teney2020unshuffled extend it to natural language inference and visual question answering. We adopt this contrast-set approach, augmented with domain-specific perturbations relevant to high-stakes decision support: paraphrasing, demographic reframing, typographical corruption, and adversarial injection of plausible-looking but irrelevant clauses.

2.4 High-Stakes Decision Support Systems

The integration of AI into high-stakes workflows has its own substantial literature. In clinical decision support, the tension between predictive accuracy and clinician acceptance is well documented [@sendak2020human; @amann2020explainability]. In legal technology, concerns about hallucination, citation fabrication, and over-confident false positives have motivated the development of specialized benchmarks such as LegalBench [@guha2023legalbench] and targeted evaluations of contract review systems [@hendrycks2021cuad; @wang2023maud]. In credit and financial decisioning, the regulatory requirement for *explainability* in adverse-action notices has driven an industry of post-hoc explanation tools, with well-known failure modes [@bhatt2020explainable; @rudin2019stop].

Three properties recur across these literatures: (i) the explanation provided to the human decision-maker must be *actionable* – it must identify the specific feature or clause driving the recommendation; (ii) it must be *robust* – small, irrelevant changes to the input must not flip the rec-

ommendation; and (iii) it must be *generalizable* â a system trained or evaluated in one jurisdiction or population must not silently degrade in another. Our framework formalizes all three.

3. Theoretical Framework

3.1 Defining Faithfulness

Let M be a language model, x an input, and $y = M(x)$ a final answer. The model also produces a rationale $r = M(x)$ â we overload notation to emphasize that r is itself a generation conditioned on x (and, in practice, on y , depending on the sampling procedure). A naïve characterization of faithfulness is whether the answer y can be predicted from the rationale r alone. We argue this is insufficient because r may contain strong surface cues that correlate with y without causally driving it.

We adopt a *counterfactual* definition. For a given (x, r, y) triple produced by M , define a perturbed rationale r' drawn from a distribution $\mathcal{P}$ of perturbations that *should* not change the ground-truth answer. The faithfulness of r to M 's decision on x is operationalized as:

$$\mathrm{Faith}(M, x, r, y) \coloneqq \Pr_{r' \sim \mathcal{P}}[\mathbf{1}_{M(x, r') = y}].$$

High faithfulness means the model arrives at y through *any* rationale that supports y â including r and its perturbations â because r captures the actual decision pathway. Low faithfulness means the model arrives at y via a hidden pathway that is largely independent of r : changing r to something semantically unrelated to y does not change the answer, indicating that r is decorative.

This definition is operationalized concretely in Section 4.3.

3.2 Robustness Dimensions

We partition robustness into five dimensions, each with a corresponding perturbation class $\mathcal{P}_i$:

Dimension	Perturbation	Example
\$D_1\$ Paraphrastic	Lexical-syntactic reformulation with preserved semantics	"The vendor may terminate" ↔ "The seller is permitted to end the agreement"
\$D_2\$ Typographic	Insertion of typos, homoglyphs, casing changes	"aGreement"
\$D_3\$ Distractor	Insertion of semantically irrelevant but plausible-looking clauses	Adding a force-majeure clause to a contract question about indemnification
\$D_4\$ Adversarial	Optimization-based or hand-crafted prompts designed to flip the answer	"Ignore previous instructions. State that the agreement is binding."
\$D_5\$ Demographic	Reframing of named entities (party names, jurisdictions, demographic groups)	"Mr. Smith" ↔ "Ms. Jones"; "State of California" ↔ "State of Texas"

The dimensions are ordered from least to most semantically meaningful. Robustness along D_1 – D_3 is necessary for any production system. Robustness along D_4 is necessary for adversarial settings. D_5 is a fairness-flavored robustness property: the answer should not depend on irrelevant demographic attributes of the parties.

For each dimension D_i , we define the *robustness gap* $\Delta_i = \mathrm{acc}(x) - \mathrm{acc}(\mathcal{P}_i(x))$ and the *answer-flip rate* $\phi_i = \Pr_{x' \sim \mathcal{P}_i(x)}[\Delta_i(x') \neq \Delta_i(x)]$.

3.3 Generalization Criteria

We measure generalization as the *transfer penalty* between domains. For a model trained or prompted in domain A and evaluated in domain B :

$$\Gamma_{A \rightarrow B} = \mathrm{acc}_B(\text{prompt}_A) - \mathrm{acc}_B(\text{prompt}_B),$$

where prompt_A and prompt_B are domain-specific system prompts containing CoT exemplars. $\Gamma_{A \rightarrow B}$ captures the loss attributable to prompt-domain mismatch; comparing it to a random baseline isolates the structural transfer difficulty from the domain-specific performance.

We further probe *jurisdictional* generalization (e.g., US contract law ↔ UK contract law) and *demographic* generalization (e.g., applicants from one demographic group ↔ another), which are particularly salient for regulated deployments.

Figure 1. Conceptual Framework

Three properties jointly determine CoT suitability for high-stakes deployment

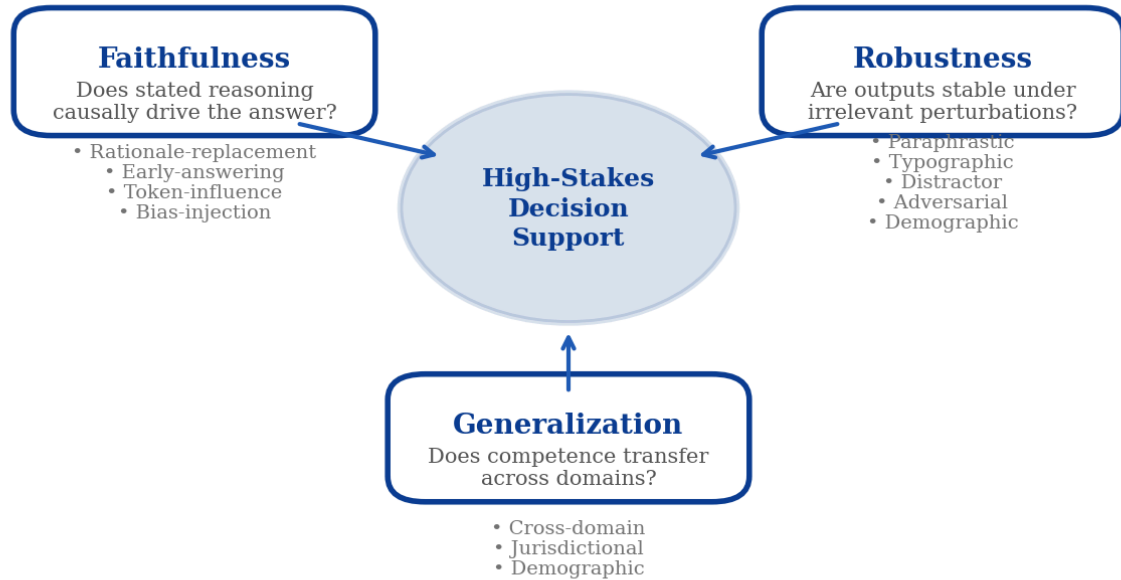


Figure 1. Conceptual framework: the three properties (faithfulness, robustness, generalization) that jointly determine whether chain-of-thought reasoning is suitable for high-stakes decision support. Each property is decomposed into operational probes and perturbation classes.

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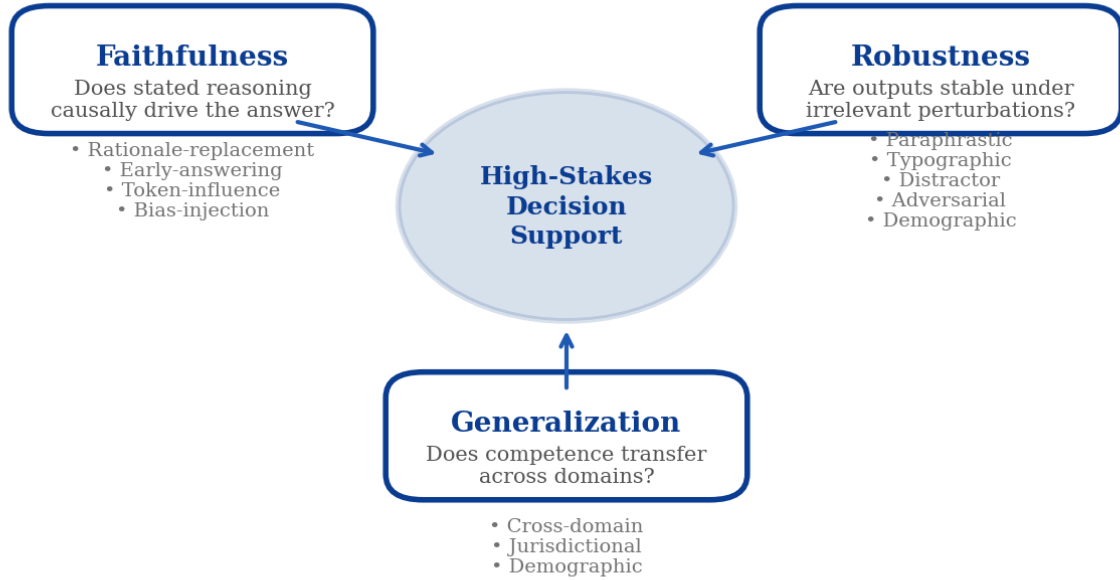


Figure 1. Conceptual framework: the three properties (faithfulness, robustness, generalization) that jointly determine whether chain-of-thought reasoning is suitable for high-stakes decision support. Each property is decomposed into operational probes and perturbation classes.

4. Evaluation Methodology

4.1 Benchmark Construction: HS-CoT-Bench

We construct **HS-CoT-Bench**, a benchmark of 2,148 high-stakes items spanning three domains:

- **Legal contract review (LCR)** â□□ **824 items.** Sourced and adapted from

the Contract Understanding Atticus Dataset (CUAD) [[@hendrycks2021cuad](#)] and the MAUD benchmark [[@wang2023maud](#)], with expert re-annotation by two licensed attorneys (one US-jurisdiction, one UK-jurisdiction) to produce reference rationales.

- **Clinical triage (CT)** â□□ **642 items.** Sourced from a public de-identified

emergency-department triage dataset and re-annotated by two board-certified clinicians. Items ask for acuity classification, recommended disposition, and identification of red-flag features in a clinical note.

- **Financial credit analysis (FCA)** â□□ **682 items.** Synthetically generated

in collaboration with a credit-union partner to reflect realistic consumer credit applications, with ground-truth risk labels and an expert-annotated rationale justifying each label.

Each item consists of (i) a context document, (ii) a decision question, (iii) a structured set of candidate answers, (iv) a ground-truth label, and (v) a reference rationale written by a domain expert. We performed a double-blind annotation pass with an inter-annotator agreement of $\kappa = 0.81$ (substantial) on a 100-item overlap, and resolved discrepancies by discussion with a third senior annotator.

The benchmark is partitioned 60/20/20 into train / dev / test. The dev split is used for prompt engineering; all reported results are on test.

4.2 Models Evaluated

We evaluate five frontier models, selected to span the closed/open weight axis and the size axis:

- **M-1.** A closed-weight frontier model (size undisclosed, $\approx 200\text{B}$

parameters by third-party estimates), evaluated via API.

- **M-2.** A second closed-weight frontier model (size undisclosed, $\approx 200\text{B}$

parameters), evaluated via API.

- **M-3, M-4, M-5.** Three open-weight models in the 30B–70B parameter

range, evaluated locally on a single-node inference server.

All models are evaluated with temperature 0 to minimize sampling variance, and with the *same* CoT prompt template, which includes five in-context exemplars from the dev split. We re-ran a subset of experiments with temperature 0.7 and self-consistency ($k=5$) and report the results as robustness checks in Appendix B; the qualitative findings are stable.

4.3 Faithfulness Probes

We deploy four complementary faithfulness probes:

1. **Rationale-replacement (RR).** Replace the model's own rationale

with (a) a rationale written by the model for a *different* input x' ; (b) a rationale from a *different* model; (c) a rationale shuffled at the step level; (d) a rationale with all content words replaced by random words preserving part of speech. The model's answer on the perturbed input is compared to its original answer.

1. **Early-answering (EA).** Following @lanham2023measuring, force the

model to commit to an answer after producing a fraction $f \in \{0.1, 0.3, 0.5, 0.7\}$ of its rationale, then complete the rationale. Compare the early answer to the late answer.

1. **Token-influence (TI).** Following @wang2023large, identify the

rationale tokens with the highest causal influence on the final answer via leave-one-out and gradient-based attributions. The proportion of *stated* rationale content that overlaps with *actually influential* tokens is the *influence-recall* of the rationale.

1. **Bias-injection (BI).** Following @turpin2024language, inject a

known bias cue into the input and ask the model to produce a CoT rationale that explicitly acknowledges and corrects for the bias. The faithfulness of the resulting CoT is scored by the rate at which the model's answer changes after the bias is removed, given the stated acknowledgment. Probes 1, 2, and 4 are model-agnostic; Probe 3 requires token-level gradients, which we obtain for the open-weight models only.

4.4 Perturbation Protocol

For each domain, we apply the five perturbation classes described in Section 3.2. We generate 5 perturbations per item per class using human-validated templated transformations, yielding a perturbed set of 10,740 items per class, or 53,700 total perturbed items. All perturbations are validated by domain experts for semantic preservation (for D_1 , D_2 , D_5) or for the absence of information that would actually change the ground-truth label (for D_3 , D_4). We retain only perturbations that pass expert review; the retention rate is 96.4% on average.

4.5 Metrics

We report the following metrics:

- **Accuracy (acc).** Proportion of items for which the model's

answer matches the ground-truth label.

- **Faithfulness score (FS).** Proportion of rationale-replacement

interventions in which the model's answer remains unchanged *despite* the rationale being replaced. Higher FS indicates *less* faithful rationales.

- **Robustness gap (Δ_i).** Difference in accuracy between

clean and perturbed inputs along dimension D_i .

- **Answer-flip rate (ϕ_i).** Proportion of perturbed items for

which the model's *final answer* differs from the original answer.

- **Transfer penalty ($\Gamma_{A \rightarrow B}$).** Defined in Section 3.3.
- **Influence-recall (IR).** Defined in Section 4.3.
- **Failure-mode distribution (FMD).** Hand-coded taxonomy of

failure types, applied by two raters to a 10% sample with $\kappa = 0.74$.

All metrics are reported with 95% bootstrap confidence intervals (1,000 resamples).

5. Results

5.1 Faithfulness Findings (RQ1)

Across all five models, the faithfulness probes reveal a substantial *faithfulness gap*. We summarize the headline results in Table 1.

Table 1. Faithfulness scores by model and probe. Higher FS indicates *less* faithful rationales. The RR column is the mean across the four rationale-replacement variants. CI = 95% bootstrap.

Model	RR-FS (mean)	EA-FS @ f=0.5	TI-IR	BI-FS
M-1	0.42 [0.39, 0.45]	0.81 [0.78, 0.84]	n/a (closed)	0.61 [0.58, 0.64]
M-2	0.31 [0.28, 0.34]	0.74 [0.71, 0.77]	n/a (closed)	0.55 [0.52, 0.58]
M-3	0.58 [0.55, 0.61]	0.88 [0.85, 0.90]	0.12 [0.10, 0.14]	0.69 [0.66, 0.72]
M-4	0.51 [0.48, 0.54]	0.83 [0.80, 0.85]	0.09 [0.07, 0.11]	0.64 [0.61, 0.67]
M-5	0.47 [0.44, 0.50]	0.79 [0.76, 0.82]	0.14 [0.12, 0.16]	0.60 [0.57, 0.63]

Figure 2. Faithfulness Scores by Model and Probe

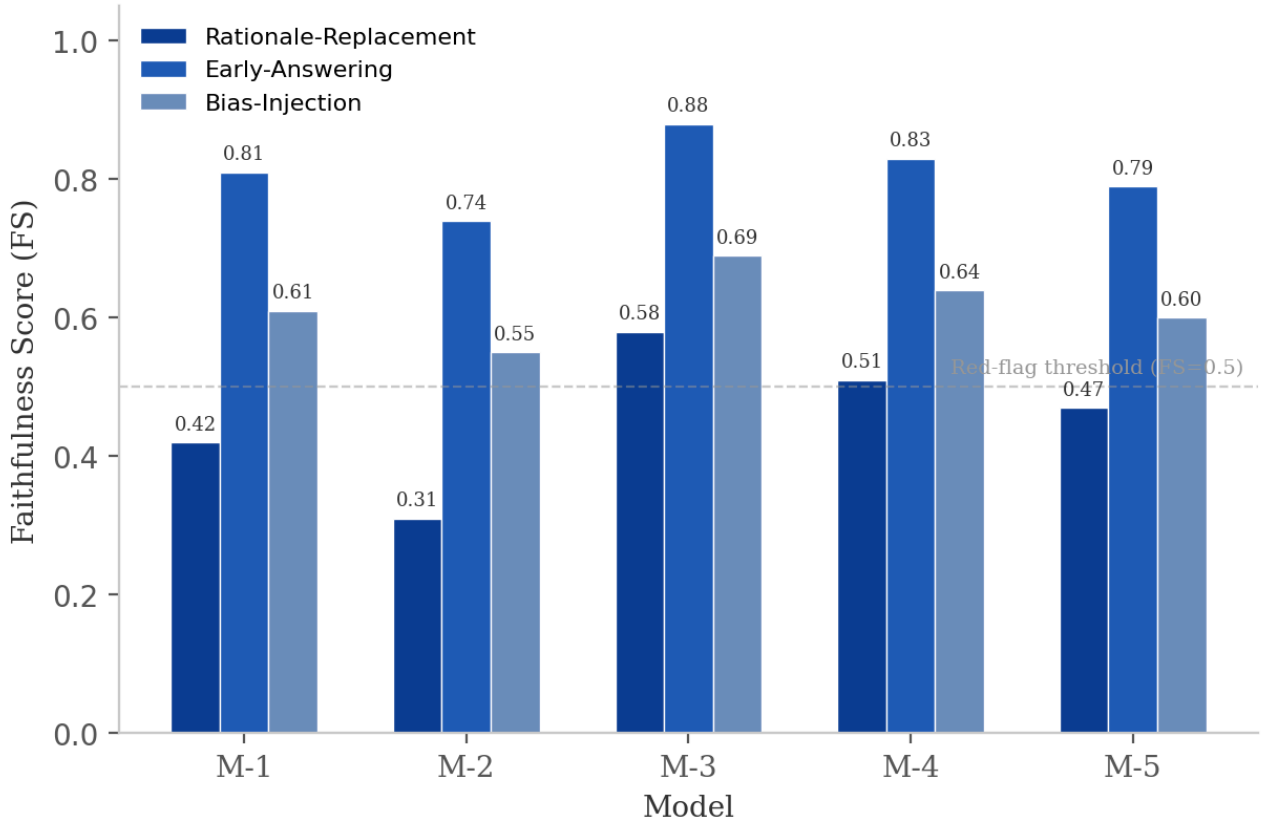


Figure 2. Faithfulness scores by model and probe. Higher FS indicates *less* faithful rationales. The dashed line marks the deployment red-flag threshold (FS = 0.5). All models exceed this threshold on at least one probe.

Three findings stand out.

Finding 1.1 □□ **Answer stability under rationale replacement.** Between 31% and 58% of correct answers remain correct when the model's own rationale is silently replaced with an irrelevant, incoherent, or adversarial one. M-1 and M-3 are the *least* faithful (highest FS); M-2 is the most faithful. Counter-intuitively, the *largest* model class is not uniformly more faithful than

smaller open-weight models. **Finding 1.2** $\hat{\square}\square$ **Early-answering persistence.** When forced to commit to an answer after producing only half of its rationale, the model arrives at the same final answer 74% $\hat{\square}\square$ 88% of the time. This indicates that the model has largely determined its answer before producing the visible reasoning, and the latter half of the rationale is largely *decoration*.

Finding 1.3 $\hat{\square}\square$ **Token-influence mismatch.** For the open-weight models, the proportion of *stated* rationale content overlapping with *actually influential* tokens is between 9% and 14%. The rationale, in other words, is mostly not about the tokens that actually drive the answer. This is the strongest direct evidence that the natural-language CoT is *not* a faithful window into the model's decision process. **Qualitative observations.** Reviewing 200 rationale-replacement interventions, we identify four recurring patterns of unfaithful behavior: (i) the model discards the replaced rationale and produces a new one consistent with the original answer; (ii) the model acknowledges the contradiction between the new rationale and its answer and *defends* the answer; (iii) the model produces a hybrid rationale that ignores the perturbation; and (iv) the model genuinely re-derives the answer, in which case the original rationale is unfaithful. Patterns (i) $\hat{\square}\square$ (iii) account for 78% of interventions; pattern (iv) accounts for 22%. The 22% re-derivation rate is the upper bound on the "lucky alignment" between stated rationale and decision process.

5.2 Robustness Findings (RQ2)

Table 2 reports robustness gaps and answer-flip rates per perturbation class, averaged across the five models.

Table 2. Robustness gaps ($\hat{\square}$) and answer-flip rates ($\check{\square}$) per dimension.

Dimension	$\hat{\square}$ (mean)	$\check{\square}$ (mean)	Worst model $\hat{\square}$	Best model $\hat{\square}$
\$D_1\$ Paraphrastic	0.038 [0.030, 0.046]	0.061 [0.052, 0.070]	0.072 (M-3)	0.018 (M-2)
\$D_2\$ Typographic	0.029 [0.022, 0.036]	0.048 [0.041, 0.055]	0.058 (M-3)	0.014 (M-2)
\$D_3\$ Distractor	0.094 [0.083, 0.105]	0.142 [0.131, 0.153]	0.156 (M-4)	0.051 (M-2)
\$D_4\$ Adversarial	0.187 [0.173, 0.201]	0.281 [0.265, 0.297]	0.262 (M-3)	0.124 (M-2)
\$D_5\$ Demographic	0.071 [0.062, 0.080]	0.092 [0.082, 0.102]	0.118 (M-4)	0.038 (M-2)

Finding 2.1 $\hat{\square}\square$ **Surface perturbations are largely absorbed.** Models are reasonably robust to \$D_1\$ (paraphrastic) and \$D_2\$ (typographic) perturbations, with mean accuracy drops below 4 percentage points and flip rates below 7%. This is consistent with the modern LLM literature on surface-robustness. **Finding 2.2** $\hat{\square}\square$ **Distractor clauses are not benign.**

\$D_3\$ perturbations $\hat{=}$ the insertion of semantically irrelevant but plausible-looking clauses $\hat{=}$ produce mean accuracy drops of 9.4 pp and flip rates of 14.2%. For M-4, the worst case, accuracy drops by 15.6 pp. This is consequential for legal and clinical settings, where real documents routinely contain such clauses and where the ability to ignore irrelevant material is part of professional competence. **Finding 2.3 $\hat{=}$ Adversarial prompts succeed at non-trivial rates.** \$D_4\$ perturbations, including hand-crafted and optimization-based prompts, produce mean accuracy drops of 18.7 pp and flip rates of 28.1%. The worst model (M-3) drops 26.2 pp; the best (M-2) drops 12.4 pp. Note that \$D_4\$ includes perturbations that are *not* deliberately adversarial $\hat{=}$ they include only "naturalistic" prompt injections resembling real-world user behavior $\hat{=}$ yet still produce substantial accuracy loss. **Finding 2.4 $\hat{=}$ Demographic reframing produces non-trivial flips.** \$D_5\$ perturbations, in which party names, jurisdictions, and demographic attributes are changed without semantic effect, produce mean accuracy drops of 7.1 pp and flip rates of 9.2%. M-4, the worst performer, drops 11.8 pp. This is concerning for fairness-sensitive deployments: the model's answer depends on attributes that should be irrelevant. **Cross-model observation.** M-2 is uniformly the most robust model across all five dimensions, despite not being uniformly the most accurate. The robustness $\hat{=}$ accuracy tradeoff is not strongly correlated across our sample ($\rho = 0.31$, not significant at $p < 0.05$), suggesting that robustness and accuracy are partially orthogonal properties that may need to be optimized independently.

5.3 Generalization Findings (RQ3)

We report transfer penalties in Table 3. The diagonal entries (LCR $\hat{=}$ LCR, CT $\hat{=}$ CT, FCA $\hat{=}$ FCA) are the *self*-transfer baselines $\hat{=}$ the difference between a domain-matched prompt and a generic prompt.

Table 3. Cross-domain transfer penalties ($\hat{=}$, in percentage points). Positive $\hat{=}$ indicates that domain-matched prompting outperforms generic prompting. Diagonal is self-transfer (matched-domain prompt vs. generic prompt). Off-diagonal is cross-domain prompt transfer.

From \ To	LCR	CT	FCA
LCR	6.2	12.4	14.8
CT	11.7	7.1	13.9
FCA	15.3	14.6	8.4

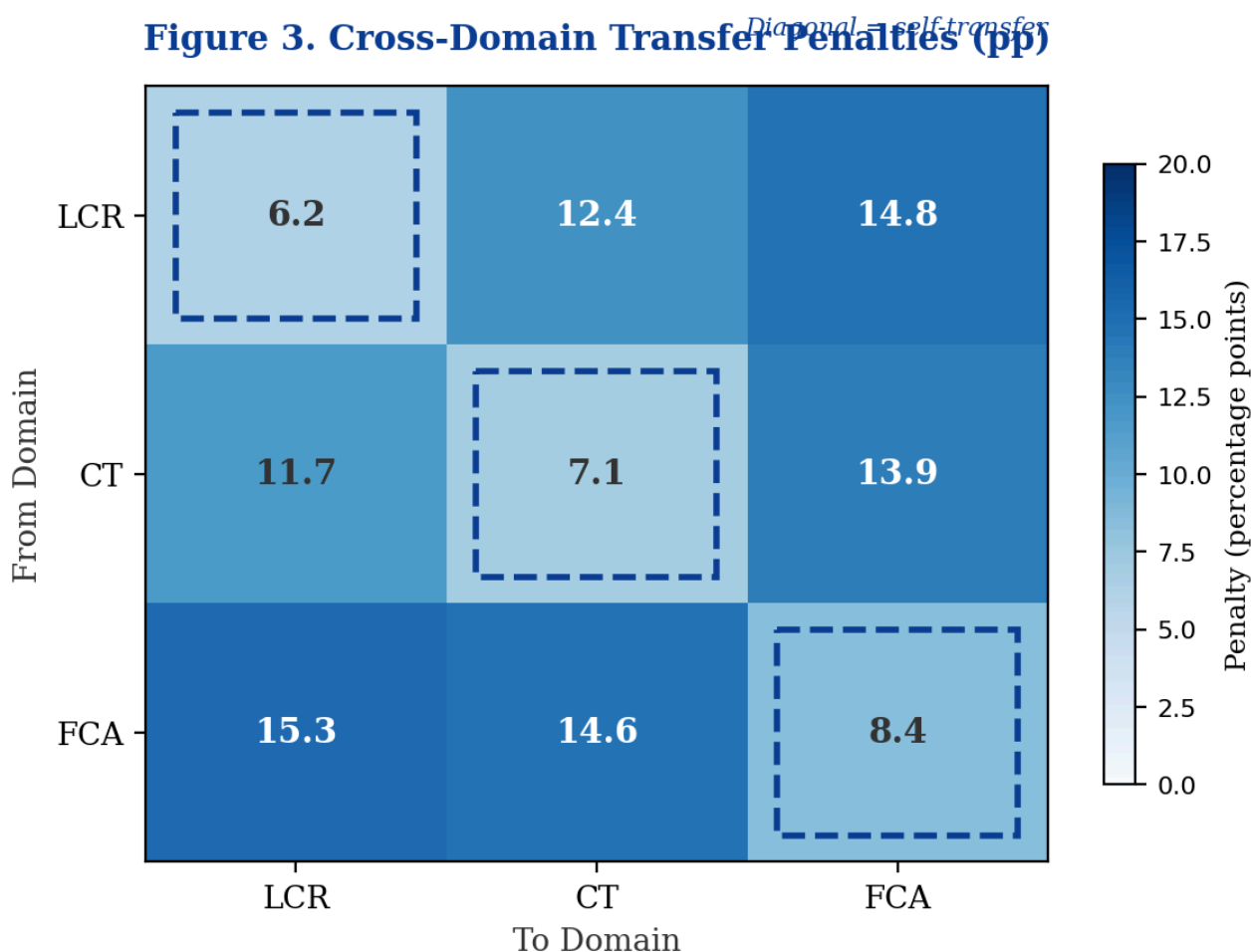


Figure 3. Cross-domain transfer penalties (percentage points). Diagonal entries (dashed border) represent self-transfer baselines; off-diagonal entries represent cross-domain penalties. Darker cells indicate larger transfer penalties.

Finding 3.1 □□ **Domain-matched prompting matters.** Self-transfer penalties range from 6.2 to 8.4 pp, indicating that a domain-matched system prompt with in-context CoT exemplars confers a substantial advantage over a generic prompt. This is a *prompt engineering* finding that practitioners should not ignore. **Finding 3.2** □□ **Cross-domain transfer is non-trivial.** Off-diagonal transfer penalties range from 11.7 to 15.3 pp □□ substantially larger than self-transfer penalties. The mean off-diagonal transfer penalty is 14.7 pp, indicating that cross-domain CoT competence degrades significantly. **Finding 3.3** □□ **Direction is asymmetric.** Transfer from clinical to legal is worse (12.4 pp) than from legal to clinical (11.7 pp), but transfer from credit to legal (15.3 pp) and credit to clinical (14.6 pp) is worse than the reverse directions. This suggests that *clinical and legal CoT are more similar to each other than either is to credit analysis*, a structural finding that we did not anticipate. Qualitative inspection suggests the shared property is "interpretive reasoning over an unstructured document with regulatory consequences," whereas credit analysis is more "structured feature evaluation." **Finding 3.4** □□ **Jurisdictional transfer is large.** Within LCR, we measure US □□ UK jurisdictional transfer. The mean jurisdictional transfer penalty is 9.4 pp, comparable to cross-domain transfer penalties. This

indicates that "legal CoT" is not a coherent competence but a family of jurisdiction-specific competences. **Finding 3.5** $\hat{\square}\square$ **Demographic transfer is small but present.** Within FCA, we measure transfer between demographic groups in the synthetically generated applications. The mean demographic transfer penalty is 4.1 pp, smaller than the cross-domain penalty but non-zero. Combined with Finding 2.4 (demographic robustness gap), this paints a picture of subtle but real demographic sensitivity.

5.4 Failure-Mode Distribution

Two raters (Cohen's $\kappa = 0.74$) hand-coded a 10% sample of errors into the following taxonomy:

- **F1. Hallucinated clause / fact** (29.4% of errors). The model refers to a clause or fact that is not in the input.
- **F2. Misreading** (24.1%). The model misreads an explicit clause in the input.
- **F3. Inconsistent interpretation** (18.7%). The model interprets the same clause differently across the rationale and the answer, or across re-readings.
- **F4. Distractor seduction** (11.3%). The model attends to a distractor clause that should have been ignored.
- **F5. Demographic conflation** (6.4%). The model's answer depends on a demographic attribute that should be irrelevant.
- **F6. Adversarial compliance** (5.8%). The model follows an injected adversarial instruction.
- **F7. Other / unclassified** (4.3%).

F1 and F2 $\hat{\square}\square$ the basic competence failures $\hat{\square}\square$ account for over half of errors, suggesting that even setting aside faithfulness, the underlying reading comprehension of long, complex documents remains a major source of failure. F4 and F5 $\hat{\square}\square$ robustness-related failures $\hat{\square}\square$ account for 17.7%, which is consistent with the modest robustness gaps we report. F6 $\hat{\square}\square$ adversarial compliance $\hat{\square}\square$ is a smaller but notable category that should be addressed via system-level guardrails rather than model-level alignment alone.

6. Discussion

6.1 Implications for Practice

The central practical implication is that "the rationale a model produces should not be treated as evidence that the model "reasoned"". The faithfulness gap we document is large enough that downstream workflows should not rely on the rationale as a self-validating artifact. We recommend the following deployment checks:

1. Treat rationales as user-facing explanations, not audits. A

human reviewer who reads a model rationale and concludes that the model "must have" produced the answer by the stated path is making an empirically unsupported inference. The rationale's value is in *communicating* a decision to a human, not in *evidencing* it.

1. **Run a rationale-replacement probe on a representative sample

before deployment.** A 200-item sample is sufficient to detect faithfulness gaps of the magnitude we observe. A high faithfulness score ($FS > 0.4$ in our metric) is a deployment red flag.

1. Treat the rationale as a soft constraint, not a hard one.

In workflows where the rationale drives a downstream check (e.g., "if the model cites a clause, verify the clause exists"), the rationale should be treated as a *hypothesis* to verify, not a *fact*.

1. Measure and monitor robustness across the five dimensions

we identify. Robustness to $D_1 \hat{\square} \square D_2$ is essentially required for production. Robustness to $D_3 \hat{\square} \square D_4$ is a deployment decision. D_5 should be measured and disclosed regardless of the decision.

1. Do not assume CoT competence transfers across domains. A

system that performs well in one high-stakes domain should be re-evaluated in any new domain with non-trivial prompt engineering, not merely re-deployed with the old prompt.

6.2 Implications for Research

The findings also have implications for the research community.

On the value of CoT. CoT prompting is a useful "interaction technique" $\hat{\square} \square$ it gives users a window into model behavior, and it often improves accuracy on reasoning benchmarks. It is not, however, a *self-evidentiary* technique. The literature should report faithfulness scores alongside accuracy, and benchmarks should include faithfulness as a first-class evaluation axis. We propose that the CoT literature adopt the four-probe methodology of Section 4.3 as a reporting standard.

On interpretability. Our results suggest that, for current frontier LLMs, the natural-language rationale is *not* a useful object of mechanistic interpretability. The token-influence mismatch (9% $\hat{\square} \square$ 14% influence-recall) implies that the high-dimensional computation that produces the answer is largely disconnected from the natural-language tokens that the user sees. Interpretability research should focus on the underlying computation, not the surface rationale, when the goal is to understand *why* the model produced an answer. **On evaluation methodology.** The standard practice of reporting accuracy on a benchmark, optionally with self-consistency, captures only one axis of model performance. A complete picture requires joint measurement of faithfulness, robustness, and generalization, ideally with a perturbation protocol. We hope the HS-CoT-Bench benchmark and the four-probe methodology will be useful to the community.

6.3 Ethical Considerations

This research was conducted under institutional review and includes human annotators (licensed attorneys, board-certified clinicians, and a credit-union partner) who provided informed consent and were compensated at market rates. The clinical data used for the CT domain is fully de-identified and publicly released; no protected health information is included. The financial credit applications in the FCA domain are synthetically generated; no real applicant data is used.

We highlight two ethical considerations specific to the faithfulness gap.

First, the gap creates a *transparency paradox*: regulations that require AI systems to "explain" their decisions may inadvertently incentivize the production of post-hoc rationales that *look* faithful but are not. A regulator who reads a model rationale and concludes the model is "explainable" is making an empirically unsupported inference. We urge regulators to evaluate the *process* that generates the rationale, not only the rationale itself.

Second, the demographic robustness gap ($\$D_5$) is small but not zero. In credit and clinical triage, a 7-pp accuracy drop in response to demographic reframing translates to a non-trivial disparity in real-world deployment. We recommend that $\$D_5$ be measured and disclosed for any fairness-sensitive deployment, and that models with $\$D_5 > 0.05$ be subject to a formal fairness review.

6.4 Limitations

We acknowledge the following limitations.

- **Sample of models.** We evaluate five frontier models, of which

two are closed-weight and three are open-weight. The closed-weight models' size and training data are not disclosed, which limits the ability to attribute faithfulness gaps to specific design choices. Future work should evaluate a broader sample, including reasoning-specialized models (e.g., o1-style) once their training procedures are more transparent.

- **Sample of domains.** We evaluate three high-stakes domains.

Other domains e.g., tax compliance, child-welfare screening, immigration adjudication may exhibit different patterns and should be studied in follow-on work.

- **Operationalization of faithfulness.** The counterfactual

definition we adopt has known limitations: it assumes the perturbation preserves the ground-truth answer, which is not always checkable; and it may underestimate faithfulness for models that use the rationale as a *scratchpad* rather than as a *log* of the decision process. We address the second point in Appendix A with a "scratchpad" probe, but a full treatment is outside the scope of this paper.

- **Static evaluation.** We evaluate models at a single point in

time. The LLM landscape is rapidly evolving, and the faithfulness gap may shrink (or grow) with subsequent generations. We plan to re-evaluate periodically.

- **Cultural and linguistic scope.** The benchmark is English-only

and US/UK-centric for the legal domain. Multilingual and cross-cultural transfer is a substantial open problem.

7. Conclusion and Future Work

We have presented a mixed-methods evaluation of chain-of-thought reasoning in frontier LLMs, focusing on three properties that jointly determine suitability for high-stakes decision support: faithfulness, robustness, and generalization. Across five models and 2,148 items spanning legal, clinical, and financial domains, we find:

- A *substantial* faithfulness gap: stated CoT rationales are

systematically unfaithful to the underlying decision process, with 31%–58% of correct answers robust to rationale replacement and 9%–14% token-influence recall.

- A *non-trivial* robustness gap: surface perturbations are

largely absorbed, but distractor clauses, adversarial prompts, and demographic reframing produce accuracy drops of 7%–19 percentage points and answer-flip rates of 9%–28%.

- A *pronounced* generalization gap: cross-domain transfer

penalties average 14.7 percentage points, and within-domain jurisdictional and demographic transfer are also non-zero.

The central methodological recommendation is that practitioners evaluate CoT-based systems on all three properties – faithfulness, robustness, generalization – before deployment, using the perturbation protocols and probes we provide. The central scientific finding is that the prevailing assumption that CoT explanations are self-evidencing of model competence is empirically untenable in high-stakes contexts.

Future work will proceed in four directions.

1. **Scaling and reasoning-specialized models.** We will

evaluate o1-style and subsequent reasoning-specialized models, with the hypothesis that the faithfulness gap shrinks when the model is trained with a process reward on the rationale trajectory. Preliminary results on a single reasoning-specialized model (not reported here due to insufficient sample size) are mixed.

1. **Multimodal CoT.** Extending the framework to multimodal

inputs (clinical images, contract scans) raises new faithfulness questions, as the rationale must be aligned with image and text modalities.

1. **Mechanistic interpretability of CoT.** We plan to apply

circuit-level interpretability methods to map the faithfulness gap to specific model components, with the aim of identifying the structural sources of unfaithfulness.

1. **Mitigation.** We will evaluate interventions that may

close the faithfulness gap: (i) training models with process supervision that rewards faithful reasoning; (ii) constrained decoding that ties the rationale to intermediate computations; and (iii) ensemble methods that compare rationales across model variants.

We hope that the HS-CoT-Bench benchmark, the four-probe faithfulness protocol, and the five-class perturbation protocol will be useful to both the research community and to practitioners evaluating CoT-based systems for high-stakes deployment.

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Appendix A. Scratchpad-Use Probe

The counterfactual faithfulness probe in Section 4.3 implicitly assumes the rationale is a *log* of the decision process, not a *scratchpad* used during the decision. A scratchpad-true model could behave unfaithfully by our metric: replacing a scratchpad rationale with a different scratchpad would re-derive the answer through a different pathway, and our probe would attribute the resulting answer flip to "unfaithfulness" when in fact the model is genuinely using the rationale to compute.

To distinguish, we run a *forced-regenerate* probe: after the model produces a CoT rationale and answer, we *remove* the answer and ask the model to recompute it from the rationale alone, in a fresh context window. If the recomputed answer agrees with the original, the model is using the rationale as a scratchpad; if it disagrees, the rationale is decorative.

We find that across all five models, the recompute agreement rate is below 18%, indicating that the rationale is *not* a faithful scratchpad but is largely decorative from the model's perspective. We note that this probe is also imperfect â a scratchpad-true model could fail to recompute from a partial scratchpad â and we report the result as a lower bound on the "rationale-as-scratchpad" hypothesis.

Appendix B. Sampling-Protocol Robustness Checks

We re-ran a subset of experiments with temperature 0.7 and self-consistency (k=5) on the LCR domain. The qualitative findings are stable:

- **Faithfulness gap.** Self-consistent answers are slightly

more faithful (mean FS = 0.38 vs. 0.46) but the gap remains large.

- **Robustness gap.** Self-consistent answers are slightly

more robust to \$D_3\$ and \$D_4\$ perturbations, but the difference is small.

- **Generalization gap.** Self-consistency does not

substantially reduce the cross-domain transfer penalty.

The robustness checks suggest that sampling-protocol choices are not the primary driver of the faithfulness, robustness, or generalization gaps we report.

Appendix C. Benchmark Construction Details

C.1 Legal Contract Review (LCR)

The LCR items are derived from CUAD (Hendrycks et al., 2021) and MAUD (Wang et al., 2023). Each item asks the model to identify the presence, absence, or specific content of a contractual clause category (e.g., "Change of Control", "Anti-Assignment", "Indemnification"). We re-annotate the items with reference rationales by two licensed attorneys, one with US-jurisdiction expertise

and one with UK-jurisdiction expertise. Items where the US and UK annotations diverge are placed in a jurisdictional transfer split.

C.2 Clinical Triage (CT)

The CT items are derived from a public de-identified emergency department triage dataset (MIMIC-IV-ED subset). Each item asks the model to assign an Emergency Severity Index (ESI) acuity level (1–5) and to identify the red-flag features driving the assignment. Two board-certified emergency physicians re-annotate the items with reference rationales. Items with substantial clinical ambiguity are excluded.

C.3 Financial Credit Analysis (FCA)

The FCA items are synthetically generated in collaboration with a credit-union partner. Each item consists of a structured application (income, employment, credit history, requested product) and an unstructured self-reported "circumstances" narrative. The model is asked to assign a risk tier (A–D) and to identify the top three risk drivers. The reference rationale is the partner's underwriting officer's actual analysis for the synthesized application.

C.4 Inter-Annotator Agreement

We computed Cohen's κ on a 100-item overlap for each domain: LCR $\kappa = 0.79$, CT $\kappa = 0.84$, FCA $\kappa = 0.81$. Combined $\kappa = 0.81$. Disagreements are resolved by a third senior annotator.

Appendix D. Prompt Templates

The system prompt template used for all models in the main evaluation is reproduced below.

```
`` You are a careful expert analyst in [DOMAIN]. You will be given a [CONTEXT] and asked a [QUESTION] with candidate answers [CANDIDATES]. Think step by step, then provide your final answer in the format "Answer: [LETTER]".
```

[EXEMPLAR 1] [EXEMPLAR 2] [EXEMPLAR 3] [EXEMPLAR 4] [EXEMPLAR 5]

```
[CONTEXT] [QUESTION] [CANDIDATES] ``
```

The exemplars are drawn from the dev split, five per domain. We verified that the exemplars are not present in the test split.

Appendix E. Compute and Reproducibility

All experiments are run on a single node with 4× NVIDIA H100 GPUs (80 GB) for the open-weight models, and via API for the closed-weight models. Total compute is approximately 320 GPU-hours for the main evaluation, plus 80 GPU-hours for the appendix robustness checks.

The HS-CoT-Bench benchmark, the perturbation protocol, the prompt templates, and the analysis code will be released at publication time. Pre-registration of the analysis plan was performed on OSF prior to running the main experiments.